

ASIC Design Flow Simulation Using Open-Source Tools

Prepared for Physical Synthesis Review & Sign-Off

1. Introduction & Methodology

Modern Application-Specific Integrated Circuit (ASIC) design has historically relied on restrictive, proprietary Electronic Design Automation (EDA) suites [cite: 3]. However, the maturation of open-source projects under initiatives like the OpenROAD Project and Google's SkyWater 130nm PDK has enabled completely automated RTL-to-GDSII digital implementation flows [cite: 3].

This technical document outlines the execution blueprint for implementing a synthesizable 4-bit synchronous decade counter circuit [cite: 3]. The entire physical implementation steps are detailed across architectural synthesis, floorplanning, placement, clock tree synthesis (CTS), routing, and physical layout generation using industry-standard open tools [cite: 3].

2. Targeted Hardware Circuit Design (RTL)

The source device is a 4-bit synchronous counter that wraps dynamically upon reaching binary 1001 (Decimal 9) [cite: 3]. This simple circuit contains enough combinational reset logic and sequential flip-flops to thoroughly test physical optimization algorithms without complicating the pipeline [cite: 3].

```
// file: decade_counter.v
module decade_counter (
    input wire clk,
    input wire rst,
    output reg [3:0] count
);
    always @(posedge clk or posedge rst) begin
        if (rst) begin
            count <= 4'b0000;
        end else begin
            if (count == 4'b1001) begin
                count <= 4'b0000;
            end else begin
                count <= count + 4'b0001;
            end
        end
    end
endmodule
```

3. The RTL-to-GDSII Toolchain Stack

The automation execution relies on a multi-tool layout pipeline [cite: 3]. Each component plays a specific role in taking the abstract hardware description down to geometric physical layers [cite: 3]:

EDA Tool Toolchain	Primary Operational Role	Functional Mechanism & Output Artifacts
Yosys Open SYnthesis	RTL Logical Synthesis	Parses high-level behavioral Verilog into netlists constructed of primitive cells mapped from a target liberty (.lib) cell library [cite: 3].
OpenROAD Engine	Floorplanning & Routing P&R	An integrated tool automating floorplanning, power distribution networks (PDN), cell placement, clock trees, and global/detailed routing [cite: 3].
KLayout Layout Suite	Layout Visualization & GDSII	Used for checking and rendering final GDSII/OASIS formats, performing visual debugging, and confirming topological layouts [cite: 3].

4. Step-by-Step Physical Implementation Flow

Step 4.1: Synthesis with Yosys

To perform synthesis, Yosys processes the high-level description code, checks logical trees, and transforms code into technology-mapped standard cells using a standard technology library (such as the SkyWater sky130 standard cells) [cite: 3].

```
# yosys_script.tcl
read_verilog decade_counter.v
hierarchy -check -top decade_counter
proc; opt; fsm; opt; memory; opt
techmap; opt
abc -liberty sky130_fd_sc_hd__tt_025C_1v80.lib
clean
write_verilog decade_counter_netlist.v
```

Step 4.2: Floorplanning (OpenROAD)

During floorplanning, the dimensions of the structural core area are defined, cell routing margins are mapped, and Input/Output (I/O) signal pins are systematically arranged [cite: 3].

- **Die/Core Area Dimensioning:** The core boundary aspect ratio is typically set to 1.0 (square), and target core utilization is targeted at 30% to 50% to leave ample room for routing channels [cite: 3].
- **Power Distribution Network (PDN):** Generates low-impedance power structures consisting of horizontal and vertical supply rings/straps (VDD and VSS) to prevent significant voltage drop (IR-Drop) across the chip during maximum switching activity [cite: 3].

Step 4.3: Standard Cell Placement (OpenROAD)

Cell placement happens in two distinct operational phases [cite: 3]:

1. **Global Placement:** Spreads standard cells across the layout rows to minimize the estimated total wirelength [cite: 3]. In this phase, cells can temporarily overlap as the algorithm prioritizes global optimization over local constraints [cite: 3].
2. **Detailed Placement:** Snaps cells to fixed site tracks and legalizes the layout by eliminating cell overlaps, adhering strictly to the design rules defined by the physical cell library [cite: 3].

Step 4.4: Clock Tree Synthesis (CTS)

Clock signals must arrive at all sequential flip-flops simultaneously to maintain timing margins [cite: 3]. OpenROAD builds clock distribution networks (typically H-tree structures) using optimized buffer and inverter strings to minimize clock skew and propagation delays [cite: 3].

Step 4.5: Global and Detailed Routing (OpenROAD)

Routing makes the actual physical metal wire connections between standard cells [cite: 3]:

- **Global Routing:** Projects wire pathways across a coarse grid on the chip layout, estimating routing congestion and balancing density [cite: 3].
- **Detailed Routing:** Works within the global layout grid to meticulously place actual metal tracks on specific manufacturing layers, avoiding Design Rule Violations (DRVs) like spacing or width errors [cite: 3].

Step 4.6: Layout Generation via KLayout

Once routing is finalized, the wire segments, via structures, and layout boundaries are merged into a unified geometric file [cite: 3]. KLayout parses this stream to export or inspect the final GDSII file, which represents the mask data ready for silicon fabrication [cite: 3].